

PROGRAMMING LANGUAGE

Definition:

A Language or a Medium which is used to instruct a computer to perform a specific and desired task is called as a programming language.

Types of Programming Language:

Three levels of programming language as follows:

1. Low level or Machine level Programming language
2. Mid-level Programming language
3. High level Programming language

Low level or Machine level Programming language:

Easily readable, understandable and executable by the machine.

Eg: Binary Language (0s and 1s)

Mid-level language:

Has instruction set called Mnemonics that is taken as input. This input is converted to machine level by a software or an additional component called Assembler. Mostly used for precise control of hardware to attain optimal performance.

Eg for Mnemonics: ADD, MOV, SUB, DIV

Eg for Mid-level language: x86, PowerPC

Eg for usage: Micro controllers in Washing machine

High level language:

Easily readable, understandable and instructible or codable by programmer. The instructions are converted to machine level with the help of two additional components or software called Compiler and Interpreter.

Eg: Java, Python, C, C++, C#, JavaScript

Procedural programming:

Procedural programming focuses on procedures or functions that operate on data to perform simple tasks. Eg: C

Object oriented programming:

Object oriented programming uses objects that combine both attributes and methods that operate on data to perform complex tasks. Code reusability is higher. Eg: Java, Python

HISTORY OF JAVA

Java is a high-level object-oriented programming language founded by James Gosling in the early 1990s. It was officially launched on 23-May-1995.

James Gosling was working at SUN microsystems. He along with his team developed Java at SUN research labs. SUN microsystems were the original creators of Java. It was acquired by Oracle Corporation in 2010. Oracle corporation is the present owner of Java programming language.

Java is made available open-source by Oracle corporation which means it can be downloaded, installed and used by anyone free of cost.

Any computer with JDK installed can execute a Java program to provide the desired output in a suitable software.

Important features of Java:

1. Class based language
2. Case sensitive language
3. Platform independent language
4. Multi-threaded language
5. Multi-domain usable language
6. Object oriented language
7. Easily codable language
8. Flexible and Secure language

Editions of Java:

Java has three main editions:

1. Java Standard Edition (J2SE)
2. Java Enterprise Edition (J2EE)
3. Java Micro Edition (J2ME)

J2SE:

Used to develop simple stand-alone applications.

Also referred as Core Java by us.

J2EE:

Used to develop complex client-server applications.

Also referred as Advanced Java by us.

J2ME:

Used to develop mobile based applications.

Also referred as Mobile Edition of Java.

STRUCTURE OF JAVA

The instructions in Java programming language are written inside the most important component called as Class. The Class has a class keyword, a ClassName followed by a class block. The class block starts with '{' and ends with '}'. These are called as 'open of block' and 'end of block' respectively. It looks as follows:

Syntax of a class:

```
class ClassName  
{  
  
}
```

This is stored in a file called as Source file. The name of the source file is same as ClassName. The name is followed by an extension called (.java) to make the computer understand that it is a file with Java program or Java code. The name of the said source file will be **ClassName.java**

Components of Source file:

Inside the Java source file, we are allowed to create three important components: class, interface and enum.

Inside Class Block:

There are two kinds of instructions that we can give inside a class block. They are:

1. Declaration Statements

- a. Variable declaration statements
- b. Method declaration statements
- c. Class/enum declarations

2. Initializers

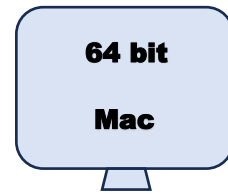
- a. Static initializers
- b. Non-Static initializers
- c. Constructors

Basic Example to understand variables and methods:

In the Login page of Facebook, you have Username, Password and Login button. The username and password text box/ text field are used to get data from user but the login button is used to perform login action.

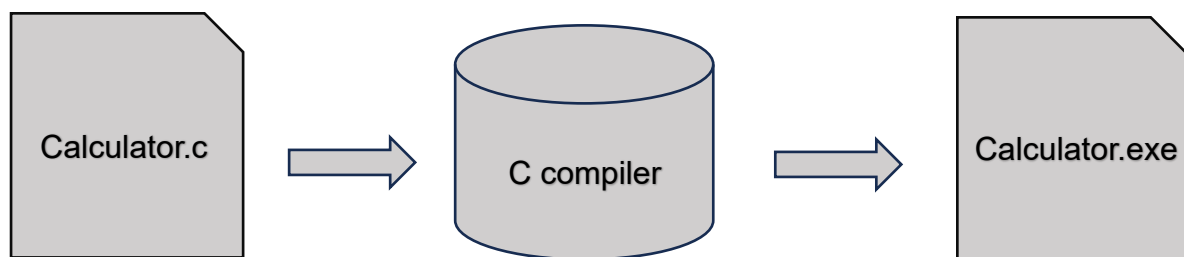
Data are stored in **Variables** and **Actions** are stored in **Methods**.

JAVA – PLATFORM INDEPENDENT / ARCHITECTURE NEUTRAL

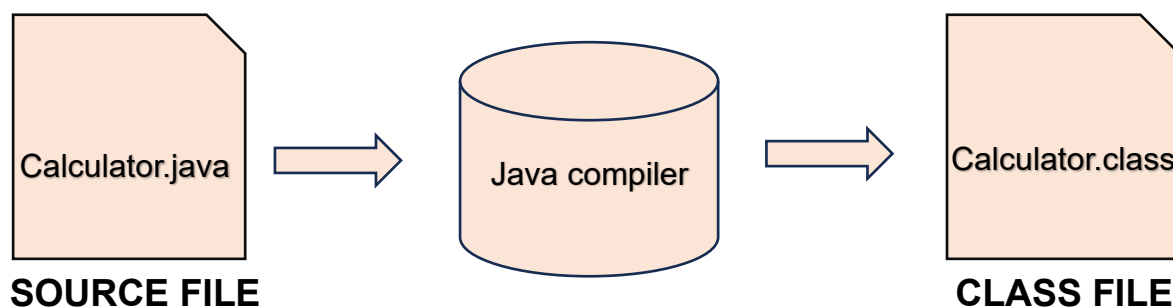


Let us consider three computers with different configurations as specified above.

A programmer wants to create a software called Calculator. At first, he chooses C programming language. He has a C compiler that is compatible with 32-bit Windows OS.



This executable file from C compiler is NOT compatible with computers with other configurations. Hence C is platform dependent programming language. Now, let us consider he chooses Java programming language.



This Class file from Java compiler can be executed by Java interpreter present in a computer of any configuration. Thus, the Java programming language offers compatibility with computers of all configurations. It can be hence called as platform independent or architecture neutral programming language.

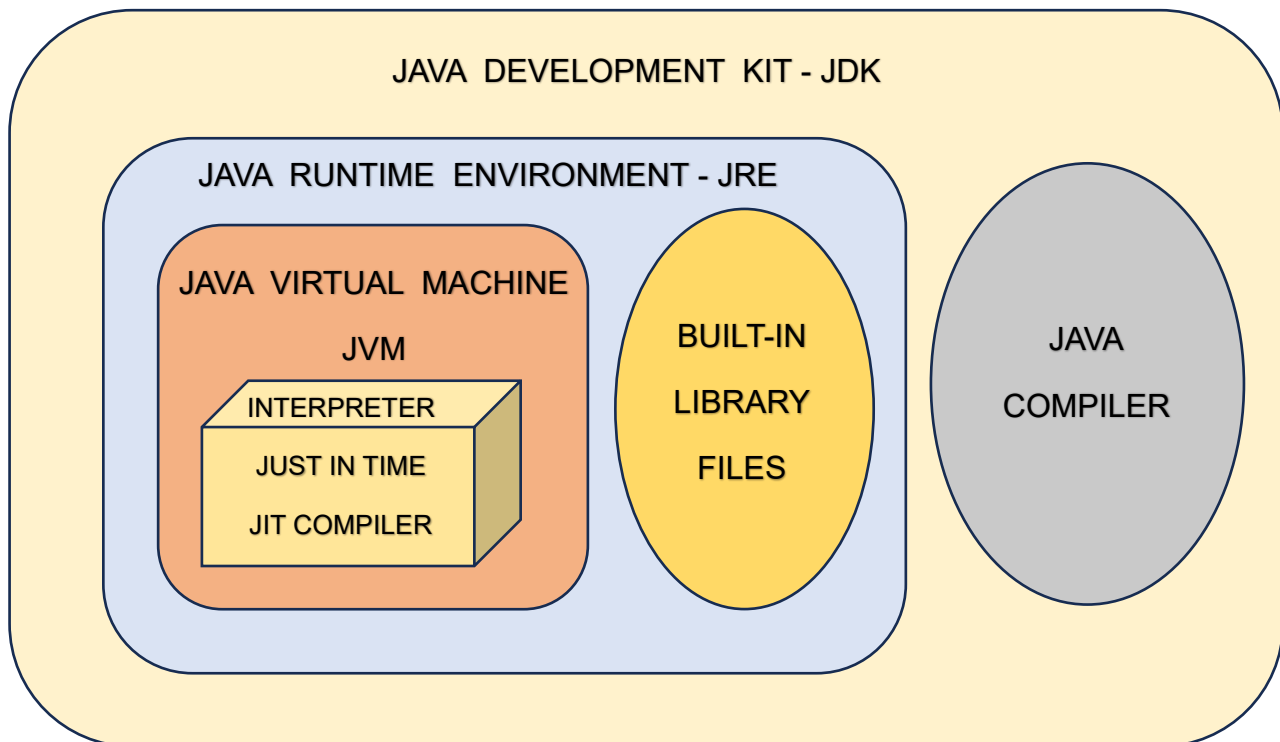
Write Once Run Anywhere (WORA):

Software developed using Java can be executed in multiple platforms. But the important pre-condition is that that computer should have JDK installed. This JDK has JRE which in turn contains JVM with Interpreter that can execute class file and give the desired output.

Hence Java is platform independent but JVM dependent language.

STRUCTURE OF JDK

Let us see how JDK looks like using a pictorial representation.



PURPOSE OF JDK

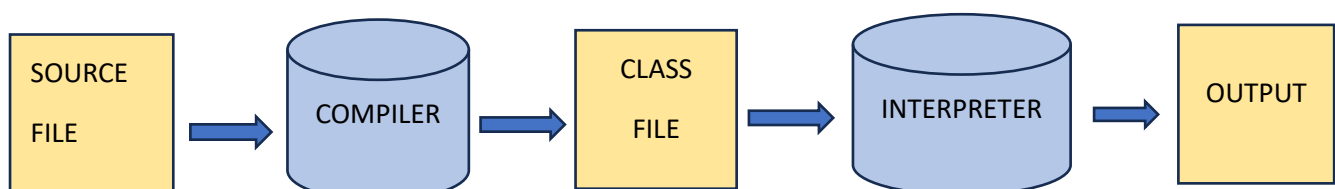
The two important components or software that are mandatory for a Java program to give the output, the Compiler and the Interpreter are both present in the JDK as seen from the above representation. Let us consider the same Calculator example to understand this. The ClassName is Calculator and so the source file name is Calculator.java

JAVA COMPILER:

The source file is converted to class file by the Java compiler with auto-generated name, Calculator.class

JAVA RUNTIME ENVIRONMENT / INTERPRETER:

The class file is also called as byte code, that is neither understood by machine, nor by programmer. That class file/byte code is converted to machine understandable form by the Interpreter present inside the JRE. Thus, the class file is taken as input by the Interpreter and final desired output is generated.



CREATE; COMPILE; EXECUTE;

STEP ONE – CREATE A JAVA PROGRAM:

The Source file can be created in a software called **Editor**. There are so many editors like Notepad, Wordpad, MS Word, Visual code and so on. It is called in short as **src file**. It has extension as (.java)

STEP TWO – COMPILE THE SOURCE FILE:

The Source file can be compiled using a software called **Compiler**. There is a Syntax or standard format that has to be used in order to compile the source file.

syntax: javac Sourcefilename.java

As soon as the syntax is written, the compilation takes place and the corresponding Class file is generated.

Rules for Successful compilation:

There should be zero syntactical errors in the program.

STEP THREE – EXECUTE THE CLASS FILE:

The class file can be executed using a software called **Interpreter**. There is a Syntax or standard format that has to be used in order to execute the class file.

syntax: java Classfilename

As soon as the syntax is written, the execution takes place and the desired output is displayed on the console screen.

Rules for Successful execution:

There should be zero logical errors in the program.

NOTE:

For steps two and three, there is a software used called Console. The best example for in-built console is Command Prompt (cmd). It is used to perform compilation and execution.

Integrated Development Environment (IDE):

The two steps,

1. Editor for scripting the program
2. Console to compile and execute the program

can be done in a single step using IDEs. The most commonly used IDE is Eclipse.

EXECUTION; PRINTING; COMMENT//

EXECUTION:

After compilation, if the program has to be executed, an important method called main method is mandatory. Because the JRE is pre-programmed in such a way that the execution starts from the main method. Method is a class member and must be declared in the class block.

Importance of Main method:

1. Start of the execution
2. Controls the flow of execution
3. End of the execution

SYNTAX OF MAIN METHOD:

```
public static void main ( String [ ] args )  
{  
}
```

PRINTING STATEMENTS:

If any message has to be printed using the Java program, printing statements are used. They must be used inside the main method. Two types:

System.out.print(data);

It is used to print the data passed on the console.

Passing data is mandatory.

System.out.println();

It is used to print the data and move the control to the next line in the console.

Passing data is optional.

COMMENT:

If there is any message to be written in the program but we do not want the compiler to read it, we can comment them.

Single line comment:

Double forward slash // is used to comment a single line.

Multi-line comment:

Slash with asterisk is used for multi-line comments. /* to start and */ to end the multi-line comment.